CREACIÓN DE CERTIFICADOS

Creación de tu PKI completa en una CA

Vamos a poner en producción una PKI completa en una sola máquina y luego distribuir los archivos según sea necesario.

1. **Instalar EasyRSA en un servidor Ubuntu**
   (mejor en una máquina virtual) y configurar los clientes que se conectarán a ese servidor:

```
sudo apt-get install easy-rsa
```

```
~ ❯ sudo apt-get install easy-rsa                                    user
[sudo] password for user:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following packages were automatically installed and are
er required:
  bridge-utils ubuntu-fan
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libccid libeac3 opensc opensc-pkcs11 pcscd
The following NEW packages will be installed:
  easy-rsa libccid libeac3 opensc opensc-pkcs11 pcscd
0 upgraded, 6 newly installed, 0 to remove and 6 not upgrade
Need to get 1.685 kB of archives.
After this operation, 5.472 kB of additional disk space will
d.
```

```
/usr/share/easy-rsa ❯ ./easyrsa --version
EasyRSA Version Information
Version:     3.2.4
Generated:   Wed Aug 27 07:28:22 CDT 2025
SSL Lib:     OpenSSL 3.5.4 30 Sep 2025 (Library: OpenSSL 3.5.4 30 Sep 2025)
Git Commit:  84be9c631c1efb872ff8cc9f16b9af4e7f018af3
Source Repo: https://github.com/OpenVPN/easy-rsa
```

2. **Inicializar tu PKI**
   Ejecuta el siguiente comando para inicializar tu PKI:

```
./easyrsa init-pki
```

```
/usr/share/easy-rsa ❯ sudo ./easyrsa init-pki
[sudo] password for kali:

Notice
------
'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:
* /usr/share/easy-rsa/pki

Using Easy-RSA configuration:
* undefined
```

3. **Crear la autoridad de certificación**
   Utiliza el siguiente comando para construir tu CA:

```
./easyrsa build-ca
```

Caso que de error

dpkg -L easy-rsa | grep /easy-rsa$

Deberías ver: /usr/share/easy-rsa

3) Crear tu carpeta de trabajo y copiar las plantillas

mkdir -p ~/easyrsa

cp -r /usr/share/easy-rsa/* ~/easyrsa/

(alternativa: crear un enlace simbólico)

ln -s /usr/share/easy-rsa ~/easyrsa

4) Trabajar desde ahí

cd ~/easyrsa

5) Inicializar la PKI

./easyrsa init-pki

6) (Opcional) comprobar versión del script

./easyrsa --version

```
/usr/share/easy-rsa ❯ sudo ./easyrsa build-ca                                    3s

Enter New CA Key Passphrase:

Confirm New CA Key Passphrase:
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...+...........+.........................+..+....+........+.+.........+...+........++++++
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [Easy-RSA CA]:loregracris

Notice
------
CA creation complete. Your new CA certificate is at:
* /usr/share/easy-rsa/pki/ca.crt

Build-ca completed successfully.
```

Puedes usar la opción nopass para desactivar el bloqueo por contraseña de la CA.

4. **Creación de certificados de servidor y cliente**
   Construir un certificado y clave de servidor: sustituye <SERVER_NAME> por el nombre de tu servidor, por ejemplo, Server01: en mi ejemplo la clave es 1234 y el servidor superalbertron

```
./easyrsa build-server-full <SERVER_NAME>
```

Puedes usar la opción nopass para desactivar el bloqueo por contraseña de la clave.

Construir un certificado y clave de cliente: sustituye <CLIENT_NAME> por el nombre de tu cliente, por ejemplo, Client01:

```
./easers' build-client-full <CLIENT_NAME>

En mi caso

./easyrsa build-client-full superalbertron

Ojo habrá que entrar primero en la carpeta easyrsa
```

```
/usr/share/easy-rsa ❯ sudo ./easyrsa build-server-full loregracris
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```

```
.....................++++++
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
-----

Notice
------
Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /usr/share/easy-rsa/pki/reqs/loregracris.req
* key: /usr/share/easy-rsa/pki/private/loregracris.key

You are about to sign the following certificate:

  Requested CN:     'loregracris'
  Requested type:   'server'
  Valid for:        '825' days


subject=
    commonName                = loregracris

Type the word 'yes' to continue, or any other input to abort.
  Confirm requested details: yes

Using configuration from /usr/share/easy-rsa/pki/e693a52e/temp.02
Enter pass phrase for /usr/share/easy-rsa/pki/private/ca.key:
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'loregracris'
Certificate is to be certified until Feb 15 06:59:34 2028 GMT (825 days)

Write out database with 1 new entries
Database updated

WARNING
=======
INCOMPLETE Inline file created:
* /usr/share/easy-rsa/pki/inline/private/loregracris.inline


Notice
------
Certificate created at:
* /usr/share/easy-rsa/pki/issued/loregracris.crt
```

Para el cliente en mi caso albertron y todas las claves 1234

Repite este paso para todos los clientes que necesites.

5. **Distribución de claves**
   Los certificados y claves del servidor y cliente deben distribuirse mediante un medio seguro, como SFTP.

6. **Descargar Easy-TLS**
   Clona o descarga Easy-TLS desde su repositorio de GitHub:
   git clone https://github.com/TinCanTech/easy-tls.git

```
~/Documents/box ❯ git clone https://github.com/TinCanTech/easy-tls.git
Cloning into 'easy-tls'...
remote: Enumerating objects: 6641, done.
remote: Counting objects: 100% (118/118), done.
remote: Compressing objects: 100% (79/79), done.
remote: Total 6641 (delta 72), reused 77 (delta 37), pack-reused 6523 (from 1)
Receiving objects: 100% (6641/6641), 5.66 MiB | 6.65 MiB/s, done.
Resolving deltas: 100% (4392/4392), done.
```

Copia Easy-TLS a tu directorio de trabajo de EasyRSA.

sudo cp easytls /usr/share/easy-rsa

7. **Inicializar Easy-TLS**
   Ejecuta los siguientes comandos:

```
./easytls init-tls

./easytls build-tls-auth

./easytls build-tls-crypt

./easytls build-tls-crypt-v2-server <SERVER_NAME>
```

```
~/easyrsa ❯ sudo ./easytls init-tls


WARNING!!!

You are about to remove the EASYTLS_PKI at: /home/kali/easyrsa/pki/easytls
and initialize a fresh TLS PKI here.

  Type the word 'yes' to continue, or any other input to abort.

  Confirm removal: yes


Saved CA Identity: /home/kali/easyrsa/pki/easytls/data/easytls-ca-identity.txt

init-tls complete; you may now create TLS keys and .inline files.
  Your newly created TLS dir is:

    /home/kali/easyrsa/pki/easytls

To configure your Easy-TLS custom group now, use:
    'easytls config custom.group YOUR_GROUP'

To configure your Easy-TLS temporary directory now, use:
    'easytls config tmp.dir YOUR_DIR'

Recommended temporary directory setting:
    *nix    - /var/easytls (Create with sudo)
    systemd - /tmp (Use systemd unit file openvpn-server@.service)
```

```
~/easyrsa ❯ sudo ./easytls build-tls-auth

TLS auth key created: /home/kali/easyrsa/pki/easytls/tls-auth.key
```

```
~/easyrsa ❯ sudo ./easytls build-tls-crypt

TLS crypt v1 key created: /home/kali/easyrsa/pki/easytls/tls-crypt.key
```

```
~/easyrsa ❯ sudo ./easytls build-tls-crypt-v2-server loregracris

TLS crypt v2 server key created: /home/kali/easyrsa/pki/easytls/loregracris-tls-crypt-v2.key
```

Crear una clave de cliente TLS-CRYPT-V2:

./easytls build-tls-crypt-v2-client <SERVER_NAME> <CLIENT_NAME>

```
~/easyrsa ❯ sudo ./easytls build-tls-crypt-v2-client loregracris cenec

TLS crypt v2 client key created: /home/kali/easyrsa/pki/easytls/cenec-tls-crypt-v2.key
```

Comprueba la carpeta donde esta esay-tls, en mi caso he tenido que hacer esto, porque he clonado dentro de la carpeta easy..

chmod +x easytls/easytls

```
# 2) (opción A) Ejecutarlo indicando la ruta

./easytls/easytls --help

./easytls/easytls init-tls

./easytls/easytls build-tls-auth

./easytls/easytls build-tls-crypt

./easytls/easytls build-tls-crypt-v2-server superalbertron

./easytls/easytls build-tls-crypt-v2-client superalbertron albertron
```

8. **Construir un archivo online**
   Dependiendo del tipo de clave TLS que estés usando (TLS auth, crypt o crypt-v2), ejecuta:

```
./easytls inline-tls-auth <SERVER_NAME>
```

Repite este paso para todos tus nodos VPN.

```
~/easyrsa ❯ sudo ./easytls inline-tls-auth loregracris

Inline TLS auth file created: /home/kali/easyrsa/pki/easytls/loregracris.inline
```

Usando la opción nokey:

```
./easytls inline-tls-auth <SERVER_NAME> --nokey
```

Finalmente hemos creado una PKI completa utilizando EasyRSA y EasyTLS, asegurando que todos los certificados y claves se generen y distribuyan de manera segura.
Esto es fundamental para establecer conexiones VPN seguras.

FIRMAR CORREOS ELECTRÓNICOS CON S/MIME USANDO THUNDERBIRD Y CERTIFICADOS PROPIOS

Instalación

Primero, necesitamos instalar las herramientas que utilizaremos para crear la CA y los certificados, para ello usaremos una máquina virtual de Ubuntu:

```
sudo apt-get update

sudo apt-get install openssl

sudo apt-get install thunderbird
```

```
~/easyrsa ❯ sudo apt-get install openssl
Reading package lists... Done
Building dependency tree... Done
```

```
~/easyrsa ❯ sudo apt-get install thunderbird
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
thunderbird is already the newest version (1:140.4.0
The following packages were automatically installed
  base58 binutils-mingw-w64-i686 binutils-mingw-w64-
```

2. Crear la autoridad de certificación (CA)

```
mkdir ~/myCA

cd ~/myCA

mkdir newcerts private

touch index.txt

echo 01 > serial
```

3. Generar la clave privada de la CA

```
openssl genpkey -algorithm RSA -out private/ca.key -pkeyopt rsa_keygen_bits:2048
```

```
~/easyrsa ❯ mkdir ~/myCA
~/easyrsa ❯ cd ~/myCA
~/myCA ❯ mkdir newcerts private
~/myCA ❯ touch index.txt
~/myCA ❯ echo 01 > serial
~/myCA ❯ openssl genpkey -algorithm RSA -out private/ca.key -pkeyopt rsa_keygen_bits:2048
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```

4. Crear el certificado de la CA

```
openssl req -key private/ca.key -new -x509 -out ca.crt -days 365 -subj
"/C=ES/ST=Valencia/L=Valencia/O=MiOrganizacion/CN=Mi CA"
```

```
~/myCA ❯ openssl req -key private/ca.key -new -x509 -out ca.crt -days 365 -subj "/C=ES/ST=Valen
cia/L=Valencia/O=MiOrganizacion/CN=Mi CA"
~/myCA ❯ cat private/ca.key
-----BEGIN PRIVATE KEY-----
MIIEvgIBADANBgkqhkiG9w0BAQEFAASCBKgwggSkAgEAAoIBAQCTiJuuTsw9aS+N
EhTSXsm2jeG6wBU2HbYbmMyY6dDtdhcofmf1ft3djW7h7FKNEZ03dBXK9x9NQBkX
p2SAxqkkrtTPdZYI36O47zDK2DhR9Puxr7Zy+/cGoeHgsmIBYcl6cv2aZxOzzRXu
w24+E6fU5luXdvLK9rxilO07h4/iZR+HE5sDjwLYEOa8h9773b9u+V+0cX9g/H+a
ViVaEO50RGXhDSHZwuM34e7Bb/uUUnVYOUfhOBI7+Dj6KHPUgVlA+21vUn/5sZ+1
F9K7pDCV0wh7ZBXcRMpVOkGjIMDq4rB7GNaw3VAzYlwIK3vMWvKD5ATBnikadKzg
lf9j9NqtAgMBAAECggEAAQkxBVDNU1MP6pQpbEQOc4cx12SuNsfk9Z2li3SngGYC
CfPqmUkvl/TRxf6pomRGGn5orBj/HG+U8tsR3sDxAsPC1Rq/bRww40VeritFffcp
```

5. Crear y firmar certificados para S/MIME

```
openssl genpkey -algorithm RSA -out private/email.key -pkeyopt rsa_keygen_bits:2048
```

```
~/myCA ❯ openssl genpkey -algorithm RSA -out private/email.key -pkeyopt rsa_keygen_bits:2048
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```

6. Crear una CSR (solicitud de firma de certificado)

```
openssl req -new -key private/email.key -out email.csr -subj "/C=ES/ST=Valencia/L=Valencia/O=MiOrganizacion/CN=miusuario"

openssl x509 -req -in email.csr -CA ca.crt -CAkey private/ca.key -CAcreateserial -out email.crt -days 365
```

```
~/myCA ❯ openssl req -new -key private/ca.key -out email.csr -subj "/C=ES/ST=Valencia/L=Valenci
a/O=MiOrganizacion/CN=miusuario"
```

```
~/myCA ❯ openssl x509 -req -in email.csr -CA ca.crt -CAkey private/ca.key -CAcreateserial -out
email.crt -days 365
Certificate request self-signature ok
subject=C=ES, ST=Valencia, L=Valencia, O=MiOrganizacion, CN=miusuario
~/myCA ❯ ls
ca.crt  ca.srl  email.crt  email.csr  index.txt  newcerts  private  serial
```

```
~/myCA ❯ openssl x509 -req -in email.csr -CA ca.crt -CAkey private/ca.key -CAcreateserial -out
email.crt -days 365
Certificate request self-signature ok
subject=C=ES, ST=Valencia, L=Valencia, O=MiOrganizacion, CN=miusuario
~/myCA ❯
```

1. Generar clave privada para email (si no existe)

openssl genrsa -out private/email.key 2048

2. Generar CSR (Certificate Signing Request)

openssl req -new -key private/email.key -out email.csr

3. Firmar el CSR con tu CA

openssl x509 -req -in email.csr -CA ca.crt -CAkey private/ca.key -CAcreateserial -out email.crt -days 365

4. Crear archivo PKCS12

openssl pkcs12 -export -out email.p12 -inkey private/email.key -in email.crt -certfile ca.crt

7. Convertir a formato PKCS#12

```
openssl pkcs12 -export -out email.p12 -inkey private/email.key -in email.crt -certfile ca.crt
```

8. Configurar Thunderbird para usar S/MIME e importar el certificado

- Abre Thunderbird.
- Ve a **Editar** > **Configuración de la cuenta**.
- En **Seguridad**, selecciona **Gestionar certificados**.
- En la pestaña **Sus Certificados**, selecciona **Importar** y elige el archivo email.p12.
- Introduce la contraseña establecida.
- Marca la opción **Firmar digitalmente los mensajes salientes**.

```
openssl pkcs12 -export -out email.p12 -inkey private/email.key -in email.crt -certfile ca.crt
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:es
State or Province Name (full name) [Some-State]:as
Locality Name (eg, city) []:asd
Organization Name (eg, company) [Internet Widgits Pty Ltd]:as
Organizational Unit Name (eg, section) []:as
Common Name (e.g. server FQDN or YOUR name) []:as
Email Address []:as

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:csubires
An optional company name []:csubires
Certificate request self-signature ok
subject=C=es, ST=as, L=asd, O=as, OU=as, CN=as, emailAddress=as
Enter Export Password:
Verifying - Enter Export Password:
```

9. Probar el envío de correos firmados

- Redacta un nuevo correo en Thunderbird.
- Envía el correo a otro destinatario o a ti mismo.
- Verifica que esté firmado correctamente con S/MIME.

10. Validar la firma del correo

Cuando recibas el correo, Thunderbird mostrará un ícono indicando que el correo está firmado.
Al hacer clic, podrás ver detalles sobre la validez del certificado.

🧩 EJERCICIO SENCILLO: Crear una VPN con certificados propios (OpenVPN + EasyRSA)

🎯 **Objetivo**

Aprender a:

- Crear una **Autoridad de Certificación (CA)**
- Emitir certificados para **servidor y clientes**
- Configurar una **VPN básica con OpenVPN**

🧰 **Requisitos**

- Máquina virtual con **Ubuntu o Kali Linux**
- Paquetes: openvpn y easy-rsa

Instálalos con:

```
sudo apt update

sudo apt install openvpn easy-rsa -y
```

```
~/myCA ❯ sudo apt install openvpn easy-rsa -y
[sudo] password for kali:
openvpn is already the newest version (2.6.15-1).
easy-rsa is already the newest version (3.2.4-1).
easy-rsa set to manually installed.
The following packages were automatically installed and are no longer required:
  base58                                  python3-flaskext.wtf
  binutils-mingw-w64-i686                 python3-flatbuffers
  binutils-mingw-w64-x86-64               python3-gevent
  gcc-mingw-w64-base                      python3-gevent-websocket
```

🔹 **1. Preparar el entorno de EasyRSA**

Creamos una carpeta de trabajo para la CA:

```
mkdir ~/easyrsa

ln -s /usr/share/easy-rsa/* ~/easyrsa/

cd ~/easyrsa
```

./easyrsa init-pki

```
init-pki hard reset failed.

EasyRSA Version Information
Version:     3.2.4
Generated:   Wed Aug 27 07:28:22 CDT 2025
SSL Lib:     OpenSSL 3.5.4 30 Sep 2025 (Library: OpenSSL 3.5.4 30 Sep 2025)
Git Commit:  84be9c631c1efb872ff8cc9f16b9af4e7f018af3
Source Repo: https://github.com/OpenVPN/easy-rsa
Host: 3.2.4 | nix | Linux | /usr/bin/zsh
~/easyrsa ❯ sudo ./easyrsa init-pki

WARNING!!!

You are about to remove the EASYRSA_PKI at:
* /home/kali/easyrsa/pki

and initialize a fresh PKI here.

Type the word 'yes' to continue, or any other input to abort.
  Confirm removal: yes


Notice
------
'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:
* /home/kali/easyrsa/pki

Using Easy-RSA configuration:
* undefined
```

🔹 **2. Crear la Autoridad de Certificación (CA)**

./easyrsa build-ca nopass

👉 Se genera:

- pki/ca.crt → certificado público de la CA
- pki/private/ca.key → clave privada de la CA

```
~/easyrsa ❯ sudo ./easyrsa build-ca nopass
........+......++++++++++++++++++++++++++++++++++++++++*..+.....+......+.+..............+.+.....
+...+............+...+..+.......+..+.+.....+.........+.........+....+..+.+...+......+.....+...
.......+........+.+.....+......+...+......+.......+.....+....+.....+...+......+............+...
....+..+....+..+....+.............+..........+.....+....+..+...............+....+..+...+.+....
.++++++++++++++++++++++++++++++++++++++*.............+.+......+...+..+.......+.....+.+......+.
....+..........+...+..+.+..+................+............+..+...+.............+............+..+
.......+...+......+......+.....+...+............+.+........+..........+.....+..........+.......
..........+...+....+...+..+.+.....+......+.........+.........+...+....+.....+..........+.......
.......+.........+...+.......+..+......+....+......+.........+........+.+......++++++
.+...+..+.+.........++++++++++++++++++++++++++++++++++++++*..............+.....+...+......+.++
++++++++++++++++++++++++++++++++++*..+...........+.+.....+.+..+.++++++
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [Easy-RSA CA]:kali

Notice
------
CA creation complete. Your new CA certificate is at:
* /home/kali/easyrsa/pki/ca.crt

Build-ca completed successfully.
```

```
~/easyrsa ❯ sudo ls -R pki
pki:
ca.crt  certs_by_serial  index.txt  issued  private  reqs  revoked  serial  vars.example

pki/certs_by_serial:

pki/issued:

pki/private:
ca.key

pki/reqs:

pki/revoked:
certs_by_serial  private_by_serial

pki/revoked/certs_by_serial:

pki/revoked/private_by_serial:
~/easyrsa ❯
```

- **3. Crear el certificado del servidor**

./easyrsa gen-req servidor01 nopass

```
~/easyrsa ❯ sudo ./easyrsa gen-req servidor01 nopass
.....++++++++++++++++++++++++++++++++++++++++*...+.+..+...+.......+.....+....+++++++++++
++++++++++++++++++*......+.+..+.......+..+......+....+..............+....+.....+...+.
.+......+.......+.....+.+..+...+....+.....+................+.....+.+...+...+........+..
+........+............+.....+...+.+.........+......+..+.......+.....+.+.....+......+.+
+.......+.....+...+..........+..+.............+...........+.........+....+...+.........
..+..................+....+...............+...+...+........+............+....+.........
...+.....+...+....+..............+.+......+...+......+...+........+...+......+....+..+.
..............+..+...+.+.....+.........+..........+.....+......+.+.....+....+..+...+...
............+.+.....+.......+.....+.+.........+..+.........+.+......+.....+....+.....+.
+........+.+.....+....+...+..+...+...+............+.........+....+......+...+..++++++
......+.........+.+.....+.......+..+...+++++++++++++++++++++++++++++++++++++++*........
.+..+..........+.....+...++++++++++++++++++++++++++++++++++++++++*...........+...+....+.
++++++
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [servidor01]:kali

Notice
------
Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /home/kali/easyrsa/pki/reqs/servidor01.req
* key: /home/kali/easyrsa/pki/private/servidor01.key
```

./easyrsa sign-req server servidor01

👉 Obtienes:

- pki/issued/servidor01.crt
- pki/private/servidor01.key

```
~/easyrsa ❯ sudo ./easyrsa sign-req server servidor01
Please check over the details shown below for accuracy. Note that this request
has not been cryptographically verified. Please be sure it came from a trusted
source or that you have verified the request checksum with the sender.
You are about to sign the following certificate:

  Requested CN:     'kali'
  Requested type:   'server'
  Valid for:        '825' days


subject=
    commonName                = kali

Type the word 'yes' to continue, or any other input to abort.
  Confirm requested details: yes

Using configuration from /home/kali/easyrsa/pki/69383019/temp.02
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'kali'
Certificate is to be certified until Feb 15 09:17:58 2028 GMT (825 days)

Write out database with 1 new entries
Database updated

WARNING
=======
INCOMPLETE Inline file created:
* /home/kali/easyrsa/pki/inline/private/servidor01.inline


Notice
------
Certificate created at:
* /home/kali/easyrsa/pki/issued/servidor01.crt
```

🔹 4. Crear el certificado del cliente

```
./easyrsa gen-req cliente01 nopass

./easyrsa sign-req client cliente01
```

👉 Obtienes:

- pki/issued/cliente01.crt
- pki/private/cliente01.key

```
~/easyrsa ❯ sudo ./easyrsa gen-req cliente01 nopass
..+..+.................++++++++++++++++++++++++++++++++++++++++*....+...++
+++++++++++++++++*..........+...+..+...+...+.......................+......+.
.........+...+...+.+...+....................+.......+.....+.+.....+.......+
.+..+.......................+...+............+........+.............++++++
.+...+.+.....+...++++++++++++++++++++++++++++++++++++++++*..+++++++++++++++
+++*........+............+...+..+...+......+..................+...+........
.+...........+...+..........+......+.....+.+...+...........++++++
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [cliente01]:lokesea

Notice
------
Private-Key and Public-Certificate-Request files created.
```

```
~/easyrsa ❯ sudo ./easyrsa sign-req client cliente01
Please check over the details shown below for accuracy. Note that this request
has not been cryptographically verified. Please be sure it came from a trusted
source or that you have verified the request checksum with the sender.
You are about to sign the following certificate:

  Requested CN:     'lokesea'
  Requested type:   'client'
  Valid for:        '825' days


subject=
    commonName                = lokesea

Type the word 'yes' to continue, or any other input to abort.
  Confirm requested details: yes

Using configuration from /home/kali/easyrsa/pki/ea636d6b/temp.02
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'lokesea'
Certificate is to be certified until Feb 15 09:19:24 2028 GMT (825 days)
```

🔹 **5. Generar parámetros Diffie-Hellman**

(Para el intercambio seguro de claves)

```
./easyrsa gen-dh
```

👉 Crea pki/dh.pem

```
~/easyrsa ❯ sudo ./easyrsa gen-dh
Generating DH parameters, 2048 bit long safe prime
.....................+.....+.......................
.......................+..........................
...................................................
.......+...........................................
.............................+..................+....
................+...............................+...
...................................................
....................+..............................
...................................................
...................................................
...................................................
...................................................
*++*++*++*++*++*++*++*++*++*++*++*++*++*++*++*++*++*+
+*++*++*++*++*++*++*++*++*++*++*++*++*++*++*++*++*++*
DH parameters appear to be ok.

Notice
------

DH parameters of size 2048 created at:
* /home/kali/easyrsa/pki/dh.pem
```

🔹 **6. Copiar certificados al servidor OpenVPN**

```
sudo cp pki/ca.crt /etc/openvpn/

sudo cp pki/issued/servidor01.crt /etc/openvpn/

sudo cp pki/private/servidor01.key /etc/openvpn/

sudo cp pki/dh.pem /etc/openvpn/
```

```
~/easyrsa ❯ sudo cp pki/ca.crt /etc/openvpn/
~/easyrsa ❯ sudo cp pki/issued/servidor01.crt /etc/openvpn/
sudo cp pki/private/servidor01.key /etc/openvpn/
sudo cp pki/dh.pem /etc/openvpn/
~/easyrsa ❯ ll /etc/openvpn
total 48
drwxr-xr-x   4 root root  4096 Nov 12 10:21 ./
drwxr-xr-x 205 root root 12288 Nov 12 07:44 ../
-rw-------   1 root root  1176 Nov 12 10:21 ca.crt
drwxr-xr-x   2 root root  4096 Apr  2  2025 client/
-rw-------   1 root root   428 Nov 12 10:21 dh.pem
drwxr-xr-x   2 root root  4096 Apr  2  2025 server/
-rw-------   1 root root  4471 Nov 12 10:21 servidor01.crt
-rw-------   1 root root  1704 Nov 12 10:21 servidor01.key
-rwxr-xr-x   1 root root  1468 Apr  2  2025 update-resolv-conf*
```

- **7. Configurar el servidor OpenVPN**

Edita el archivo:

```
sudo nano /etc/openvpn/server.conf
```

Pega esto:

```
port 1194

proto udp

dev tun

ca ca.crt

cert servidor01.crt

key servidor01.key

dh dh.pem

server 10.8.0.0 255.255.255.0

keepalive 10 120

persist-key

persist-tun

user nobody

group nogroup

verb 3
```

Guarda con **CTRL+O**, **Enter**, y **CTRL+X**.

```
~/easyrsa ❯ sudo cat /etc/openvpn/server.conf
port 1194
proto udp
dev tun
ca ca.crt
cert servidor01.crt
key servidor01.key
dh dh.pem
server 10.8.0.0 255.255.255.0
keepalive 10 120
persist-key
persist-tun
user nobody
group nogroup
verb 3x
```

🔹 8. Iniciar el servidor

```
sudo systemctl start openvpn@server

sudo systemctl enable openvpn@server
```

```
~/easyrsa ❯ sudo systemctl start openvpn@server
sudo systemctl enable openvpn@server
Created symlink '/etc/systemd/system/multi-user.target.wants/openvpn@server.service' → '/usr/li
b/systemd/system/openvpn@.service'.
```

🔹 9. Configurar el cliente

Copia los archivos al cliente:

- ca.crt
- cliente01.crt
- cliente01.key

Crea el archivo de configuración:

```
sudo nano /etc/openvpn/cliente01.ovpn
```

Pega:

```
client

dev tun
```

```
proto udp
remote <IP_del_servidor> 1194
resolv-retry infinite
nobind
persist-key
persist-tun
ca ca.crt
cert cliente01.crt
key cliente01.key
verb 3
```

```
ca.key                                  cliente01.key
root@kali-virtualbox:/mnt/shared/HACER# cp ca.key /etc/openvpn/
root@kali-virtualbox:/mnt/shared/HACER# cp cliente01.crt /etc/openvpn/
root@kali-virtualbox:/mnt/shared/HACER# cp cliente01.key /etc/openvpn/
root@kali-virtualbox:/mnt/shared/HACER#
```

- **10. Conectar el cliente**

```
sudo openvpn --config /etc/openvpn/cliente01.ovpn
```

```
cat: /etc/openvpn/client01.ovpn: No such file or directory (os error 2)
kali@kali-virtualbox:/etc/openvpn$ sudo cat /etc/openvpn/cliente01.ovpn
client
dev tun
proto udp
remote 192.168.0.10 1194
resolv-retry infinite
nobind
persist-key
persist-tun
ca ca.crt
cert cliente01.crt
key cliente01.key
verb 3
```

Deberías ver algo como:

Initialization Sequence Completed

Y ya tienes el túnel VPN operativo 🛰️

```
4
2025-11-12 10:51:01 Socket Buffers: R=[212992->212992] S=[212992->212992]
2025-11-12 10:51:01 UDPv4 link local: (not bound)
2025-11-12 10:51:01 UDPv4 link remote: [AF_INET]192.168.0.10:1194
2025-11-12 10:51:01 TLS: Initial packet from [AF_INET]192.168.0.10:1194, sid=f9961acf 1a684052
2025-11-12 10:51:01 VERIFY OK: depth=1, CN=kali
2025-11-12 10:51:01 VERIFY OK: depth=0, CN=kali
2025-11-12 10:51:01 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certi
ficate: 2048 bits RSA, signature: RSA-SHA256
2025-11-12 10:51:01 [kali] Peer Connection Initiated with [AF_INET]192.168.0.10:1194
2025-11-12 10:51:01 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
2025-11-12 10:51:01 TLS: tls_multi_process: initial untrusted session promoted to trusted
2025-11-12 10:51:01 PUSH: Received control message: 'PUSH_REPLY,route 10.8.0.1,topology net30,p
ing 10,ping-restart 120,ifconfig 10.8.0.6 10.8.0.5,peer-id 0,cipher AES-256-GCM,protocol-flags
cc-exit tls-ekm dyn-tls-crypt,tun-mtu 1500'
2025-11-12 10:51:01 OPTIONS IMPORT: --ifconfig/up options modified
2025-11-12 10:51:01 OPTIONS IMPORT: route options modified
2025-11-12 10:51:01 OPTIONS IMPORT: tun-mtu set to 1500
2025-11-12 10:51:01 net_route_v4_best_gw query: dst 0.0.0.0
2025-11-12 10:51:01 net_route_v4_best_gw result: via 192.168.0.1 dev enp0s3
2025-11-12 10:51:01 ROUTE_GATEWAY 192.168.0.1/255.255.254.0 IFACE=enp0s3 HWADDR=08:00:27:40:88:
cd
2025-11-12 10:51:02 TUN/TAP device tun0 opened
2025-11-12 10:51:02 net_iface_mtu_set: mtu 1500 for tun0
2025-11-12 10:51:02 net_iface_up: set tun0 up
2025-11-12 10:51:02 net_addr_ptp_v4_add: 10.8.0.6 peer 10.8.0.5 dev tun0
2025-11-12 10:51:02 net_route_v4_add: 10.8.0.1/32 via 10.8.0.5 dev [NULL] table 0 metric -1
2025-11-12 10:51:02 Initialization Sequence Completed
2025-11-12 10:51:02 Data Channel: cipher 'AES-256-GCM', peer-id: 0
2025-11-12 10:51:02 Timers: ping 10, ping-restart 120
2025-11-12 10:51:02 Protocol options: protocol-flags cc-exit tls-ekm dyn-tls-crypt
```

```
3: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_cod
el state UNKNOWN group default qlen 500
    link/none
    inet 10.8.0.6 peer 10.8.0.5/32 scope global tun0
       valid_lft forever preferred_lft forever
    inet6 fe80::6fb8:1200:7f8f:f5d5/64 scope link stable-privacy proto k
ernel_ll
       valid_lft forever preferred_lft forever
root@kali-virtualbox:/mnt/shared/HACER#
```

```
root@kali-virtualbox:/mnt/shared/HACER# ping 10.8.0.1
PING 10.8.0.1 (10.8.0.1) 56(84) bytes of data.
64 bytes from 10.8.0.1: icmp_seq=1 ttl=64 time=1.38 ms
64 bytes from 10.8.0.1: icmp_seq=2 ttl=64 time=1.28 ms
64 bytes from 10.8.0.1: icmp_seq=3 ttl=64 time=1.68 ms
64 bytes from 10.8.0.1: icmp_seq=4 ttl=64 time=1.22 ms
64 bytes from 10.8.0.1: icmp_seq=5 ttl=64 time=1.27 ms
^C
--- 10.8.0.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 1.215/1.367/1.684/0.167 ms
root@kali-virtualbox:/mnt/shared/HACER#
```

Crear servidor y clientes OpenVPN versión libro (optativo)

1. **Servidor y Cliente: preparación del entorno**

```
sudo apt-get install openvpn easy-rsa

mkdir ~/easyrsa

ln -s /usr/share/easy-rsa/* ~/easyrsa

sudo chown kali ~/easyrsa

chmod 700 ~/easyrsa

cd ~/easyrsa
```

```
┌──(kali㉿kali)-[~/easyrsa]
└─$ sudo chown kali ~/easyrsa
chmod 700 ~/easyrsa
cd ~/easyrsa
```

2. **Servidor: inicialización de la PKI**

```
./easyrsa init-pki
```

```
Notice
------
'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:
* /home/kali/easyrsa/pki

Using Easy-RSA configuration:
* undefined
```

3. **Servidor: crear la autoridad de certificación (CA) he usado la clave 1234**

```
./easyrsa build-ca
```

```
$ ./easyrsa build-ca

ter New CA Key Passphrase:

nfirm New CA Key Passphrase:

syRSA version 3.2.4

ror

ssphrase mismatch!
```

4. **Servidor: generación de la clave y solicitud de certificado del servidor. He creado el servidor superalbertron**

   ./easyrsa gen-req superalbertron nopass

   Le he puesto la contraseña 1234

./easyrsa gen-req server nopass

```
┌──(kali㉿kali)-[~/easyrsa]
└─$ ./easyrsa gen-req server nopass
......+......+......+.+..+......+......+.+.....+...+....+...+...+.........
...+......+.+...+...+.....+.......+...+...+..+....++++++++++++++++++++++++
++++
....+.....+.+.............+.+......+......+...+...........+.+..+.........
.......+......+.........+......+.+...+...........+.+............+..+.+...+
.++++++++++++++++++++++++++++++++++++++*....+.+.....++++++

ou are about to be asked to enter information that will be incorporated
nto your certificate request.
hat you are about to enter is what is called a Distinguished Name or a DN.
here are quite a few fields but you can leave some blank
or some fields there will be a default value,
f you enter '.', the field will be left blank.

ommon Name (eg: your user, host, or server name) [server]:superalbertron

otice

rivate-Key and Public-Certificate-Request files created.
our files are:
 req: /home/kali/easyrsa/pki/reqs/server.req
 key: /home/kali/easyrsa/pki/private/server.key
```

5. **Servidor: firmar el certificado del servidor**

./easyrsa sign-req server superalbertron

```
subject=
    commonName                = superalbertron

Type the word 'yes' to continue, or any other input to abort.
  Confirm requested details: yes

Using configuration from /home/kali/easyrsa/pki/820a7387/temp.02
Enter pass phrase for /home/kali/easyrsa/pki/private/ca.key:
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'superalbertron'
Certificate is to be certified until Feb 14 12:06:35 2028 GMT (825 da

Write out database with 1 new entries
Database updated

WARNING
═══════
INCOMPLETE Inline file created:
* /home/kali/easyrsa/pki/inline/private/superalbertron.inline


Notice
──────
Certificate created at:
* /home/kali/easyrsa/pki/issued/superalbertron.crt
```

6. **Cliente: inicialización de la PKI en cada cliente**

```
./easyrsa init-pki
```

```
┌──(kali㉿kali)-[~/easyrsa]
└─$ ./easyrsa init-pki

ARNING!!!

ou are about to remove the EASYRSA_PKI at:
 /home/kali/easyrsa/pki

nd initialize a fresh PKI here.

ype the word 'yes' to continue, or any other input to abort.
 Confirm removal: yes


otice
------

init-pki' complete; you may now create a CA or requests.

our newly created PKI dir is:
 /home/kali/easyrsa/pki

sing Easy-RSA configuration:
 undefined
```

7. **Cliente: generar la clave y solicitud de certificado del cliente**

```
./easyrsa gen-req client01 nopass

.....+.+........+.......+........++++++++++++++++++++++++++++++++++++++++*..+.
....+.....+....+.........+.....+.+..............+.+.....+.......+..+.+.......
.+.....+.+...+..+.........+.+.....+.............+...+........+....+......+...
...+......+...+..+...+.......+........+...........+.+...+.........+..+......
++++++
.+....+..+...+.......+........+........+........+.......+...+.+......+++++++++++++
.............+...+.+......+...+......+..+....+...+......+.....+......+....+..
...+...+............+.+............+..+.+............+.....+........+......+
.+..+......+.......+...........+.....+...+.......+......+.....++++++
-----
ou are about to be asked to enter information that will be incorporated
nto your certificate request.
hat you are about to enter is what is called a Distinguished Name or a DN.
here are quite a few fields but you can leave some blank
or some fields there will be a default value,
f you enter '.', the field will be left blank.
-----
ommon Name (eg: your user, host, or server name) [client01]:superalbertron

otice
------

rivate-Key and Public-Certificate-Request files created.
our files are:
 req: /home/kali/easyrsa/pki/reqs/client01.req
 key: /home/kali/easyrsa/pki/private/client01.key
```

8. **Transferir la solicitud del cliente al servidor**

```
scp /path/to/client01.req server_user@server_ip:/path/to/server_pki/reqs/
```

9. **Servidor: importar la solicitud del cliente y firmarla**

```
./easyrsa import-req /path/to/server_pki/reqs/client01.req client01
./easyrsa sign-req client client01
```

10. **Transferir el certificado firmado de vuelta al cliente**

```
scp /path/to/server_pki/issued/client01.crt server_user@client_ip:/path/to/client01/
```

11. **Servidor: generar parámetros Diffie-Hellman**

```
./easyrsa gen-dh
```

12. **Servidor: copiar certificados a OpenVPN**

```
sudo cp ~/easyrsa/pki/ca.crt /etc/openvpn/
sudo cp ~/easyrsa/pki/issued/server.crt /etc/openvpn/
sudo cp ~/easyrsa/pki/private/server.key /etc/openvpn/
sudo cp ~/easyrsa/pki/dh.pem /etc/openvpn/
sudo chmod 644 /etc/openvpn/ca.crt /etc/openvpn/dh.pem /etc/openvpn/server.crt
sudo chmod 600 /etc/openvpn/server.key
```

13. **Cliente: copiar certificados a OpenVPN**

```
sudo cp ~/easyrsa/pki/client01.crt /etc/openvpn/
sudo cp ~/easyrsa/pki/private/client01.key /etc/openvpn/
sudo chmod 644 /etc/openvpn/client01.crt
sudo chmod 600 /etc/openvpn/client01.key
```

14. **Servidor: configurar OpenVPN**

```
sudo nano /etc/openvpn/server.conf
Añade las siguientes líneas:
port 1194
proto udp
dev tun
ca ca.crt
cert server.crt
key server.key
dh dh.pem
tls-server
```

```
cipher AES-256-CBC
keepalive 10 120
persist-key
persist-tun
user nobody
group nogroup
status /var/log/openvpn-status.log
log /var/log/openvpn.log
verb 3
```

Guarda los cambios y asegúrate de tener los certificados y claves en el directorio correcto.

15. Cliente: configuración del cliente OpenVPN

Edita o crea el archivo de configuración del servidor VPN /etc/openvpn/client01.ovpn y asegúrate de incluir las siguientes líneas:

Para abrirlo puedes usar el comando:

```
sudo nano /etc/openvpn/client01.ovpn
```

Introduce:

```
client
dev tun
proto udp
remote server_ip 1194      # Reemplaza “server_ip” con la IP o nombre de dominio de tu servidor
resolv-retry infinite
persist-key
persist-tun

ca ca.crt                  # Debe ser el mismo CA que usa el servidor
cert client01.crt          # Certificado del cliente
key client01.key           # Llave del cliente
```

```
data-ciphers AES-256-GCM:AES-128-GCM
data-ciphers-fallback AES-256-CBC
tls-client          # Añade esta línea para habilitar el modo cliente TLS
verb 3              # Nivel de detalle del log (ajustable)
```

Guarda los cambios y asegúrate de tener los certificados y claves en el directorio correcto.

---

16. Servidor: iniciar el servidor OpenVPN

```
sudo systemctl start openvpn@server
```

(Figura 2.35. Servidor OpenVPN ejecutándose en Ubuntu)

---

17. Servidor: iniciar el servicio OpenVPN en el servidor

Inicia el servicio OpenVPN en el servidor.

Se puede forzar la carga de la configuración con:

```
sudo openvpn --config /etc/openvpn/server.conf
```

---

18. Cliente: conectar el cliente al servidor

En el cliente, inicia la conexión con OpenVPN utilizando el archivo .ovpn:

```
sudo openvpn --config /path/to/client01.ovpn
```

---

19. Verificación

Prueba la conectividad mediante un ping desde el cliente al servidor.
Utiliza **Wireshark** para verificar que el tráfico esté cifrado.